Advanced Machine Learning, ETH Zurich

Convolutional filters

Carlos Cotrini

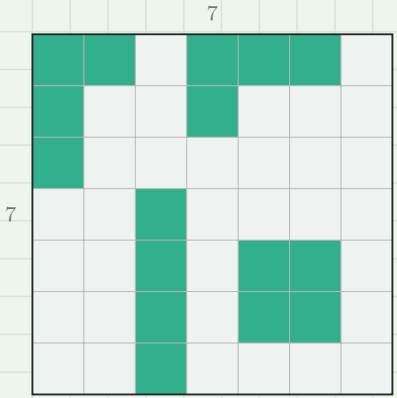
Autumn semester 2026

Subject

Decomposing an image into tiles

Entry

1

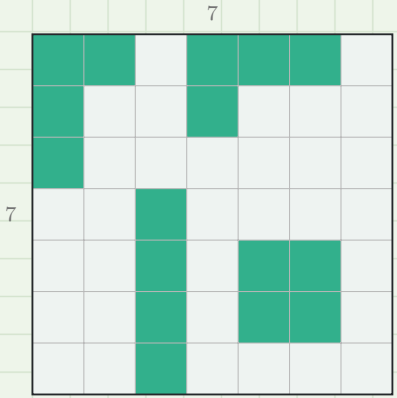


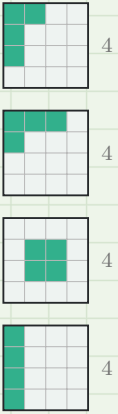
Subject

Decomposing an image into tiles

Entry

1



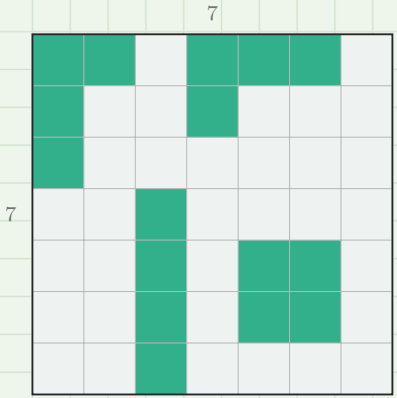


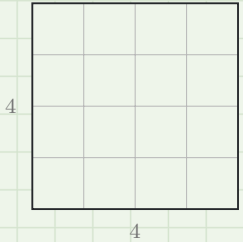
Subject

Decomposing an image into tiles

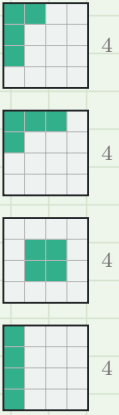
Entry

1





Feature map

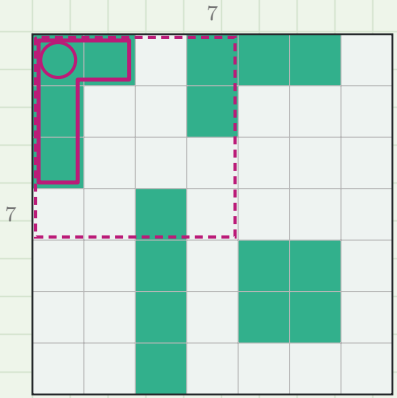


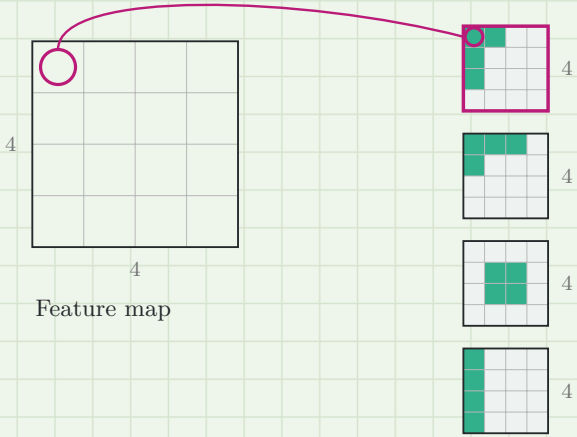
Subject

Decomposing an image into tiles

Entry

1





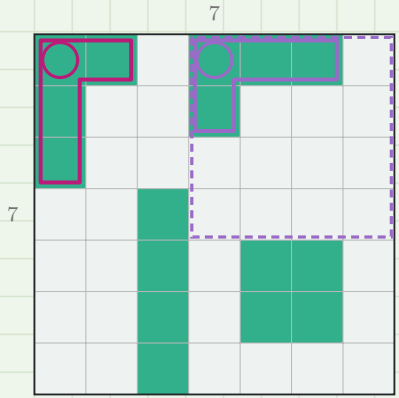
Feature map

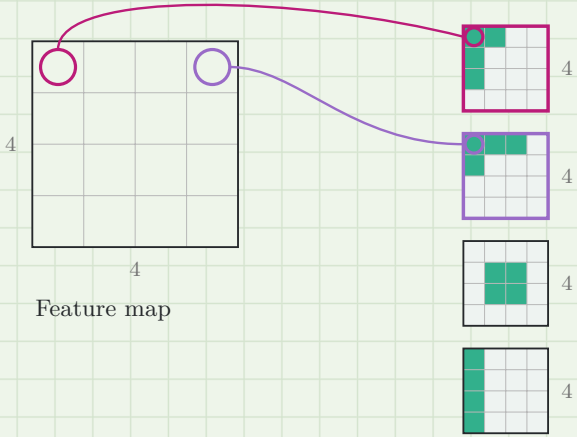
Subject

Decomposing an image into tiles

Entry

1





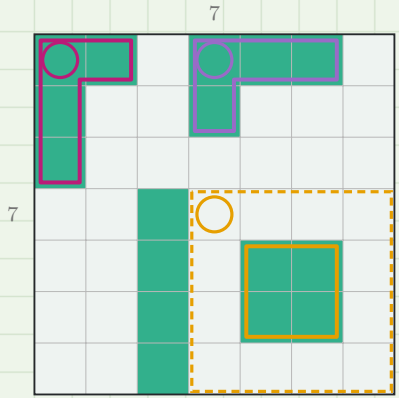
Feature map

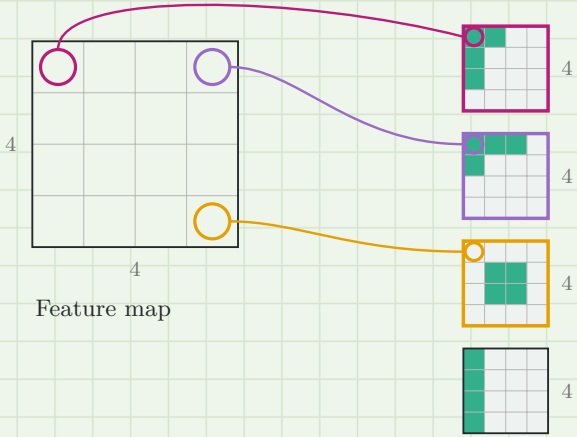
Subject

Decomposing an image into tiles

Entry

1





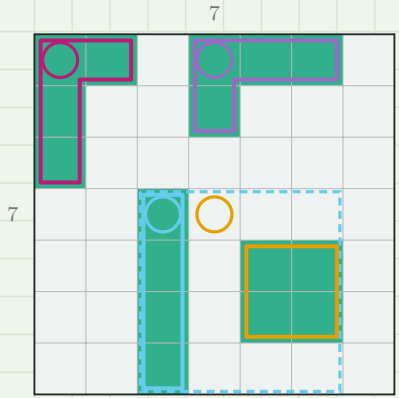
Feature map

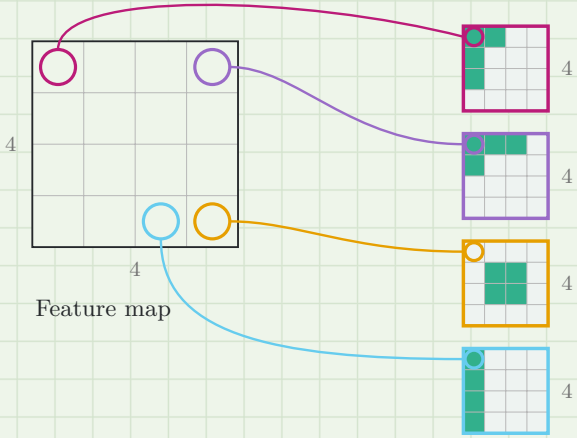
Subject

Decomposing an image into tiles

Entry

1



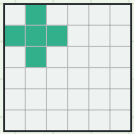


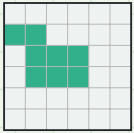
Subject

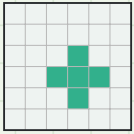
Recognizing islands that look like

Entry

2







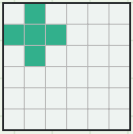
Subject

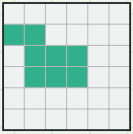
Recognizing islands that look like

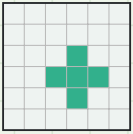
Entry

2







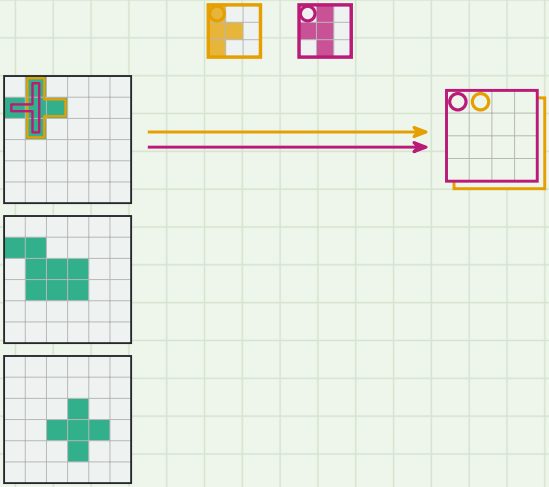


Subject

Recognizing islands that look like ✚

Entry

2

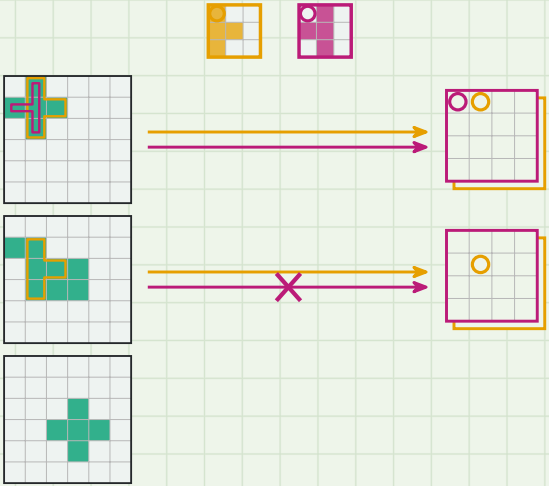


Subject

Recognizing islands that look like ✚

Entry

2

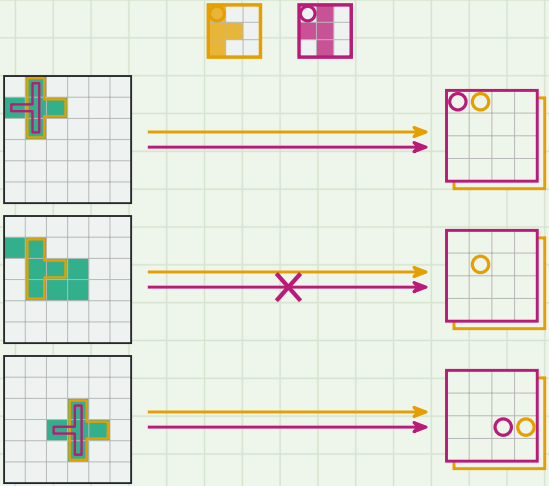


Subject

Recognizing islands that look like ✚

Entry

2



Subject

Recognizing islands that look like ✚

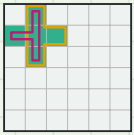
Entry

2

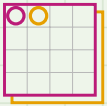


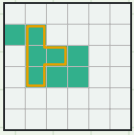


$1 \times 2 \times 2$

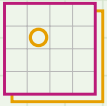


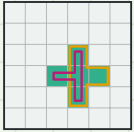




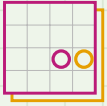












Subject

Recognizing islands that look like

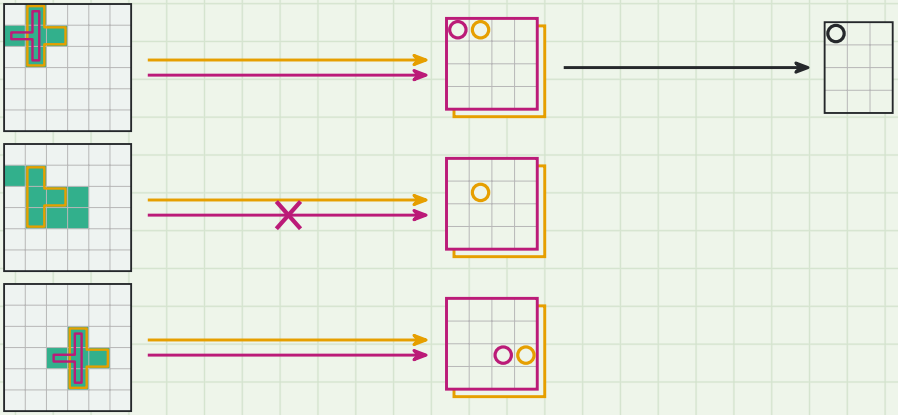
Entry

2





$1 \times 2 \times 2$



Subject

Recognizing islands that look like

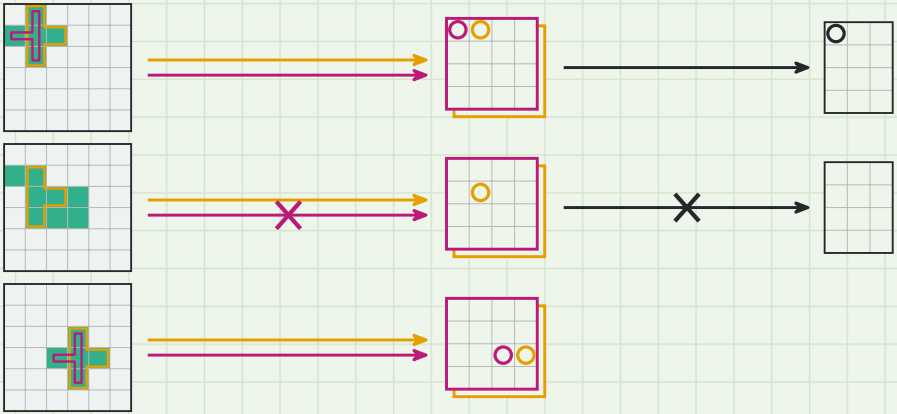
Entry

2





$1 \times 2 \times 2$



Subject

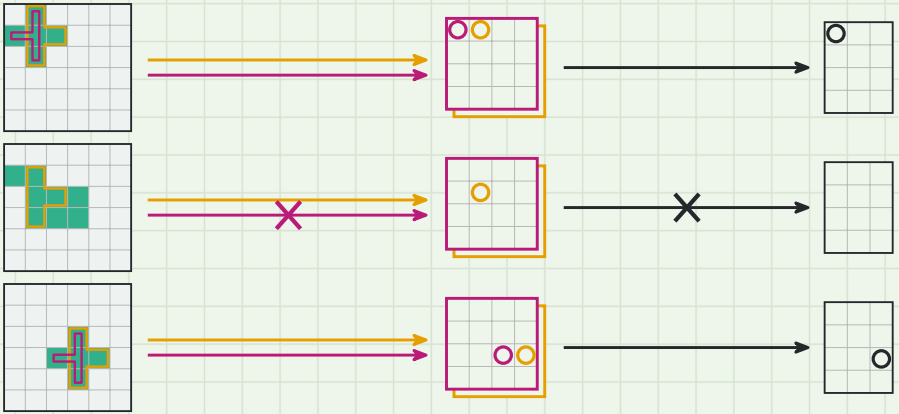
Recognizing islands that look like ✚

Entry

2









$= 1$

$= 0$

How to compute activations



= 1

= 0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	1	1	0
0	0	0	1	0	0
0	0	0	0	0	0

How to compute activations



= 1

= 0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	1	1	0
0	0	0	1	0	0
0	0	0	0	0	0



1	0	0
1	1	0
1	0	0

3×3

How to compute activations



= 1

= 0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	1	1	0
0	0	0	1	0	0
0	0	0	0	0	0



1	0	0
1	1	0
1	0	0

3×3

0	0	0		1	0	0		0	0	0	sum	0
0	0	0	$*$	1	1	0	$=$	0	0	0	$\longrightarrow$	
0	0	0		1	0	0		0	0	0		

Convolution

0			



= 1

= 0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	1	1	0
0	0	0	1	0	0
0	0	0	0	0	0



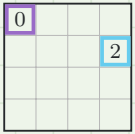
1	0	0
1	1	0
1	0	0

3×3

0	0	0		1	0	0		0	0	0	sum	0
0	0	0	$*$	1	1	0	$=$	0	0	0		
0	0	0		1	0	0		0	0	0		

0	0	0		1	0	0		0	0	0	sum	2
1	0	0	$*$	1	1	0	$=$	1	0	0		
1	1	0		1	0	0		1	0	0		

Convolution





= 1

= 0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	1	1	0
0	0	0	1	0	0
0	0	0	0	0	0



1	0	0
1	1	0
1	0	0

3×3

0	0	0		1	0	0	=	0	0	0	sum	→	0	
0	0	0	*	1	1	0		0	0	0				
0	0	0		1	0	0		0	0	0				
0	0	0		1	0	0	=	0	0	0	sum	→	2	
1	0	0	*	1	1	0		1	0	0				
1	1	0		1	0	0		1	0	0				
1	0	0		1	0	0	=	1	0	0	sum	→	4	
1	1	0	*	1	1	0		1	1	0				
1	0	0		1	0	0		1	0	0				

Convolution

0		
		2
		4



= 1

= 0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	1	1	0
0	0	0	1	0	0
0	0	0	0	0	0



1	0	0
1	1	0
1	0	0

3×3

0	0	0		1	0	0		0	0	0	sum	0
0	0	0	*	1	1	0	=	0	0	0	→	
0	0	0		1	0	0		0	0	0		
0	0	0		1	0	0		0	0	0	sum	2
1	0	0	*	1	1	0	=	1	0	0	→	
1	1	0		1	0	0		1	0	0		
1	0	0		1	0	0		1	0	0	sum	4
1	1	0	*	1	1	0	=	1	1	0	→	
1	0	0		1	0	0		1	0	0		

Convolution

0	0	0	1
0	0	2	2
0	1	2	**4**
0	0	2	2



= 1

= 0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	1	1	0
0	0	0	1	0	0
0	0	0	0	0	0



1	0	0
1	1	0
1	0	0

3×3

0	0	0		1	0	0		0	0	0	sum	0
0	0	0	$*$	1	1	0	$=$	0	0	0	→	
0	0	0		1	0	0		0	0	0		

0	0	0		1	0	0		0	0	0	sum	2
1	0	0	$*$	1	1	0	$=$	1	0	0	→	
1	1	0		1	0	0		1	0	0		

1	0	0		1	0	0		1	0	0	sum	4
1	1	0	$*$	1	1	0	$=$	1	1	0	→	
1	0	0		1	0	0		1	0	0		

Convolution

0	0	0	1
0	0	2	2
0	1	2	**4**
0	0	2	2

-3 →

−3	−3	−3	−2
−3	−3	−1	−1
−3	−2	−1	1
−3	−3	−1	−1



0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	1	1	0
0	0	0	1	0	0
0	0	0	0	0	0



1	0	0
1	1	0
1	0	0

3×3

0	0	0		1	0	0		0	0	0	sum	0
0	0	0	$*$	1	1	0	$=$	0	0	0	$\rightarrow$	
0	0	0		1	0	0		0	0	0		

0	0	0		1	0	0		0	0	0	sum	2
1	0	0	$*$	1	1	0	$=$	1	0	0	$\rightarrow$	
1	1	0		1	0	0		1	0	0		

1	0	0		1	0	0		1	0	0	sum	4
1	1	0	$*$	1	1	0	$=$	1	1	0	$\rightarrow$	
1	0	0		1	0	0		1	0	0		

Convolution

0	0	0	1
0	0	2	2
0	1	2	4
0	0	2	2

$\xrightarrow{-3}$

−3	−3	−3	−2
−3	−3	−1	−1
−3	−2	−1	1
−3	−3	−1	−1

$\xrightarrow{\text{ReLU}}$

Desired output

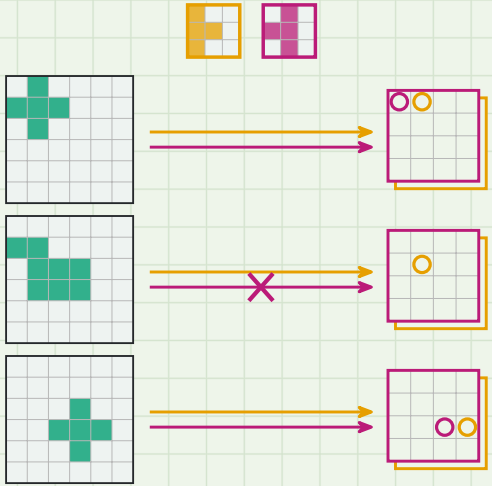
0	0	0	0
0	0	0	0
0	0	0	1
0	0	0	0

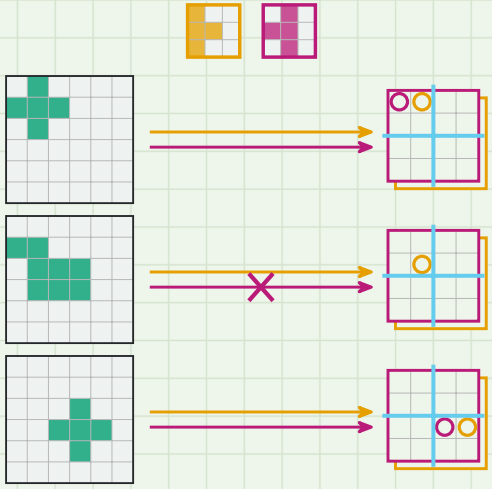
Subject

Recognizing islands that look like ✚, with pooling

Entry

4



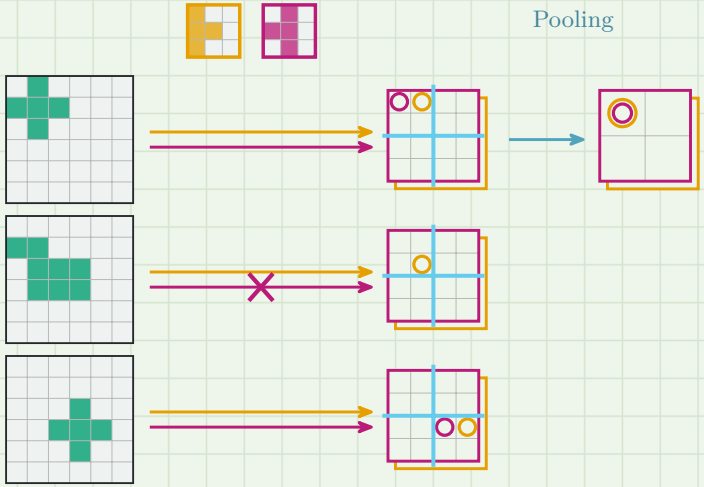

Pooling

Subject

Recognizing islands that look like ✚, with pooling

Entry

4

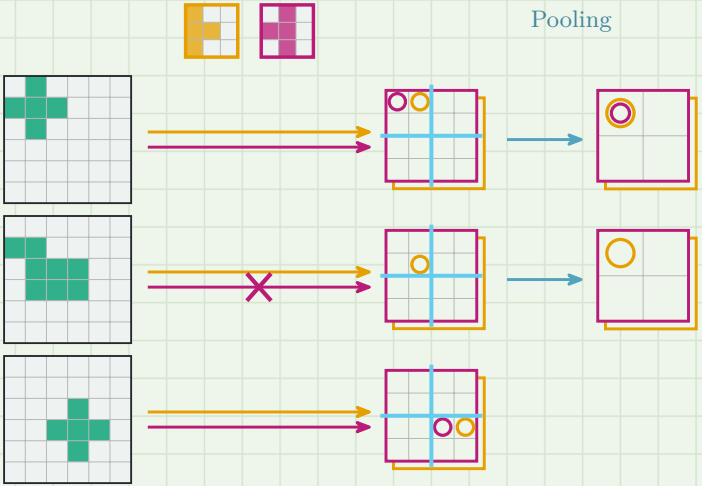


Subject

Recognizing islands that look like ✚, with pooling

Entry

4

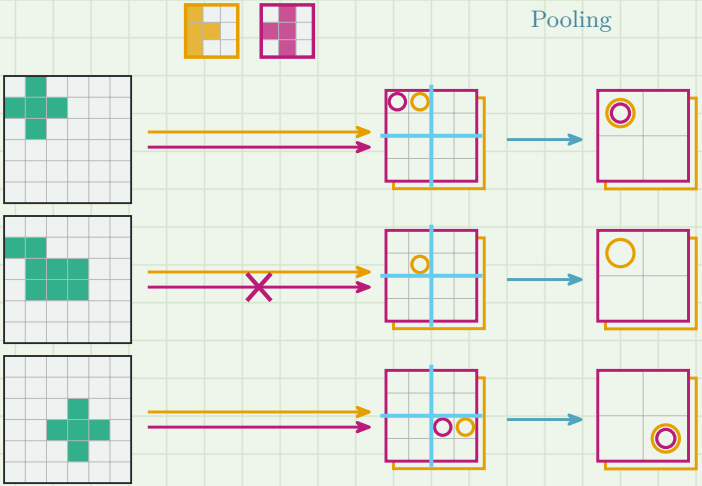


Subject

Recognizing islands that look like ✚, with pooling

Entry

4

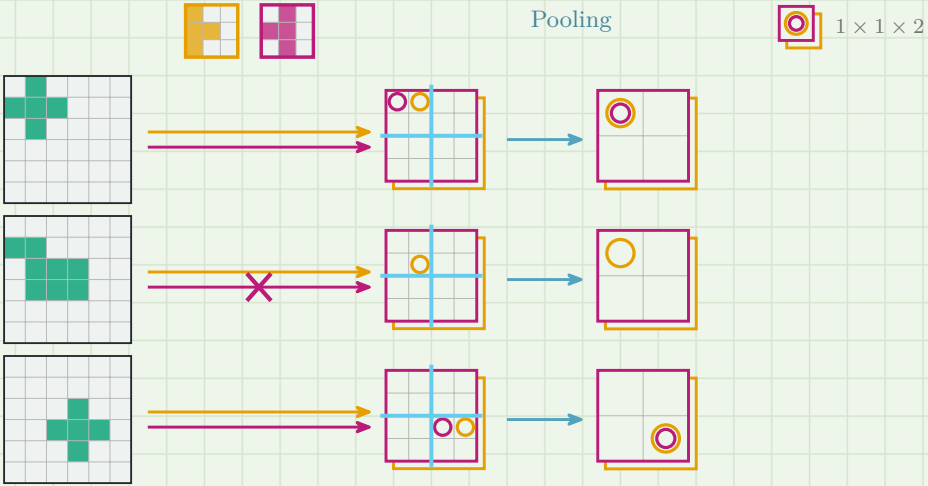


Subject

Recognizing islands that look like ✚, with pooling

Entry

4

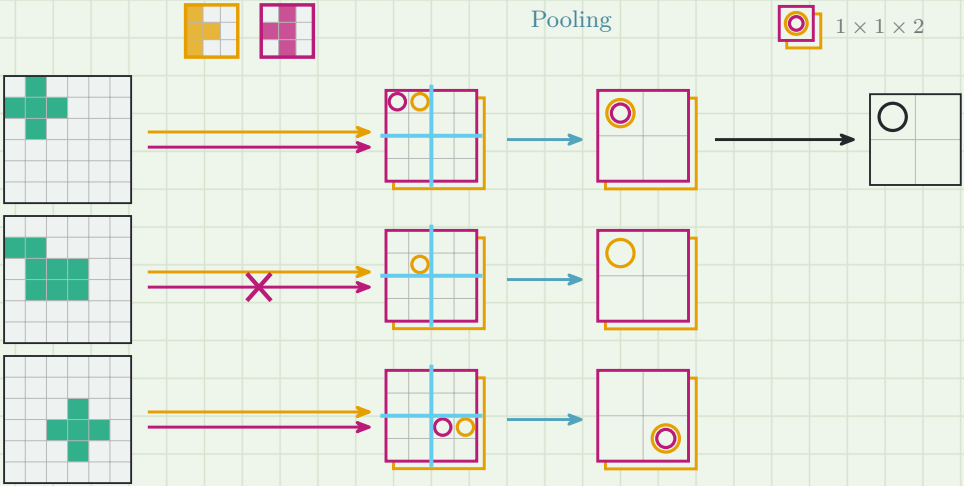


Subject

Recognizing islands that look like ✚, with pooling

Entry

4

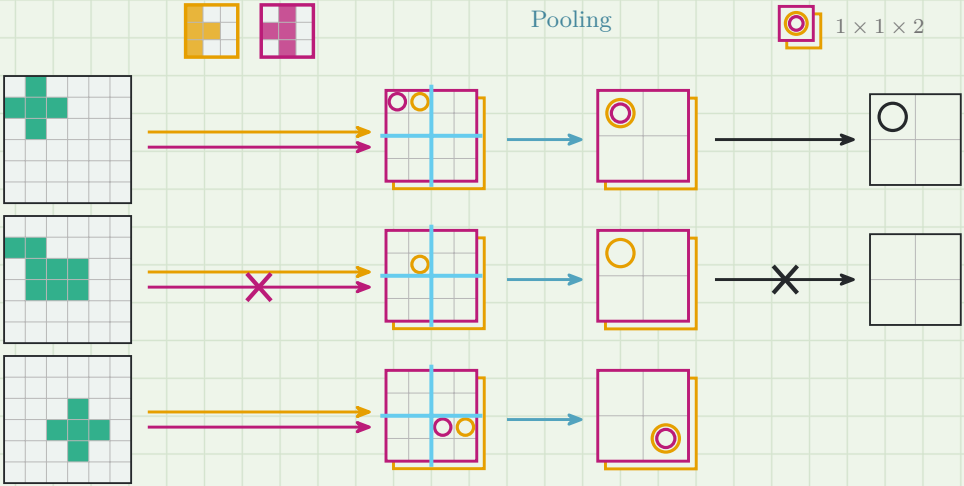


Subject

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4



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4

